Sextant

A price feed that tells you what it costs to corrupt it.

An oracle for Portaldot. Bonded reporters, a Rust median that slashes liars, and a Chainlink-compatible feed any Solidity app can read.

THE PROBLEM

Every DeFi app on Portaldot needs prices. None of them can get one they can trust.

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general price oracles among the 68 builds in this hackathon

Hardcoded

how lending, stablecoin and RWA builds get prices today, or an admin key sets them

Unknown

how much an attacker must spend to fake a price, so nobody can size their risk

HOW IT WORKS

One transaction crosses both of Portaldot's VMs.

OFF-CHAIN

Reporters

Bond tPOT. Post BTC/USD from Coinbase, Kraken, Bitstamp, Gemini, Bitfinex.

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SOLIDITY

SextantCore

Rounds close at 2/3 of stake. Slashes, pays rewards, holds bonds.

⇄

INK! · RUST

SextantMath

Stake-weighted median, MAD, outlier mask.

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SOLIDITY

SextantFeed

Chainlink `AggregatorV3Interface`. Existing code reads it unchanged.

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CONSUMER

MiniLend

Caps total debt at `corruptionCost()`. Rejects prices older than 15 min.

Rust does the maths, Solidity holds the money and speaks the interface every DeFi developer already knows.

WHAT AN ATTACKER GETS

Three outcomes, all enforced on-chain.

1 of 5 lies



Price holds

The median ignores the liar. The liar loses 20% of their bond. Honest reporters get paid.

2 of 5 collude



Feed freezes

Over 1/3 of the round disagrees. Nothing is published and no honest reporter is slashed.

4/9 of stake



That's the price

This much bonded capital can force a false price. The contract publishes the number as `corruptionCost()`.

WHY PORTALDOT

This design needs both VMs. Portaldot V3 is where they share one transaction.

ink! for the maths. Sorting and weighted medians over 16 reporters are cheap and memory-safe in Rust, at 10.4 KB of contract code.

Solidity for compatibility. Aave- or Compound-style code ports with zero oracle changes.

POT is the security budget. Every feed's safety is denominated in bonded POT, so feeds that matter create staking demand.

It unblocks the ecosystem. Lending, stablecoin, RWA and DEX builds here all need a price they can defend.

PROOF IT WORKS

Live on PortaldotV3 testnet.

SextantMath `ink!`	`0x5bc4f7bed4064425c0115dd190628dec5f8f6865` `tx 0x82908ae49d21df90e0c9711532c5cce2a2349bd5e6ef4a5cf07912e8c92b5b2c`
Cross-VM call	Solidity ABI → ink! returned median 77,555.00, MAD 5.00 and mask `0b10`: the one fake price, flagged.
Full stack v1	Core, Feed and MiniLend deployed at blocks 266,724–266,727. Reporters joined and bonded. This run exposed a PolkaVM stack overflow, which is now fixed and redeployed.
Tests	9 Forge tests (liar, collusion, whale, unbonding, lending cap, TWAP) and 5 ink! unit tests, all passing.
Dashboard	Live chain view, reporter bonds, slash log, and an Attack Lab with 5 scenarios.

BUSINESS MODEL

Apps pay per feed. Reporters earn for staking honestly.

Feed fees

Each feed is funded in POT by the apps that read it, and every honest report is paid from that pool.

Priced security

An app that needs more safety funds more stake. Its `corruptionCost()` rises, and its lending cap rises with it.

Premium feeds

FX, commodities and RWA attestations such as invoice or assay data, served through the same interface.

NEXT

Roadmap

- **Nov** Live rounds on V3 with slash and freeze demos
- **Dec** Native "5..." reporter accounts, permissionless joining
- **Q1** ETH, DOT and POT feeds, plus the first partner integration
- **Q2** Audit, then mainnet with an independent reporter set

Every lending, stablecoin and RWA app on Portaldot needs this piece. Sextant is that piece, and it says out loud how safe it is.

github.com/codeswithroh/portaldot-hackathon-2026-sextant-codeswithroh